P515/1
PRINCIPLES AND PRACTICES
OF AGRICULTURE
PAPER ONE
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3 Hours

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Uganda Advanced Certificate of Education

PRINCIPLES AND PRACTICES OF AGRICULTURE

Paper 1

TIME: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

- *This paper contains **three** sections **A,B** and **C***
- *Section **A** is compulsory*
- *Answer only **one** item from each section **B** and **C***
- *Answers to the items should be written in the answer booklets provided.*

SECTION A

VALUE ADDITION AND BASIC SCIENCE

ITEM 1

Karuru is the proprietor of seed bleedings company Ltd, an agricultural crop produce located in Kahoro trading Centre. On 10th Feb, 2026, he delivered produce to Kahoro Harembe High school. The crop products included 100Kg of beans, 180Kg of maize flour, and 1300Kg of rice, 750Kg of ground nuts and 500Kg of millet grains.

The school complained of millet grains mixed with stones and Black jack seeds, Beans of different colours and size since the school had ordered for beans of uniform size and colour.

Karuru asked if he can provide more produce subsequently since he had quantities of unsold produce in the store. Unfortunately, the school had to buy the produce once because it had other producers who would supply the same produce/commodities to the school. On his way back to the store, he made a huge loss because he was involved in a motor accident and it was crushed beyond repair. He requested someone to help him with a truck for transporting the produce to other clients,

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and promised him that he will pay him when the school pay for the supply he had made. His request did not work out because the owner of the truck needed cash.

Karuru went to one of financial institutions for a loan and was requested to bring a document to act as evidence that he has been transacting with the school for the supply of food commodities. Karuru did not have the document.

Task

a) Prepare an appropriate document to act as an evidence that shows Karuru had transactions with the school.
b) With reason, how can you advise Kakuru to improve marketing of his produce.

ITEM 2

Zamzam planted 10 hectares of Aromatic supa Rice using pure, certified seed obtained from approved seed bleeding company.

Land preparation and planting were completed at the right time, crop management practices were done uniformly across the entire field in accordance with the advice of the area agronomist.

During the growing season, some section of the field retained standing water for a long time, while another section drained rapidly and remained comparatively dry. Differences in crop growth and performance were observed between the two sections.

At maturity, some panicles were grey while others were purple, although only grey coloured seeds with grey particles were planted. Researchers collected data from representative plants in each section.

The measurements and yield records are presented below.

Table showing field data collected by researchers

Parameter	**Section A**	**Section B**
Root density	12 Roots/cm^3	6 Roots/cm^3
Stem density	148 Stems/cm^3	11 stems/cm^3
Leaf colour	Dark green	Palegreen
yield	6.lt/ha	4.1t/ha

Task

Explain how the difference in the field observations influence the rice yields at Zaaamzam's farm.

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SECTION B
ANIMAL PRODUCTION

ITEM 3

Kasaana is a goats farmer in mubende district, He started with 12 goats but he expanded the farm with another group of 8 goats that he bought from the animal market and immediately introduced them into already existing herd. He allowed his goats to feed from his 100mx50m of land fenced with barbed wire of 2(two) strands and posts.

Kasaana constructed a goat structure (house) at 15cm high with one opening as entrance. He serves his goats observed on heat two days ago, before going for work daily, he releases the goats to browse at 7:00am. Some goats are often seen licking soil, eating pieces of wood and clothes. Some goats were observed with extended stomach on the left side, unable to walk with their long toes.

Kassana planted elephant grass, kikuyu grass for his goats.

Slaughtering of goats would be done evening after browsing on bare flat surface. Kasaana got very low returns.(Adult goats should be stocked at a rate of 10-15 per hectare for optimum production)

Task

Suggest ways Kasaana's project would be improved for better yields.

ITEM 4

Janet is a poultry farmer who has specialized in rearing broilers. She bought 1300 chicks and supplied 50 round feeders and 100 drinkers in the poultry house which 10mx10m, she also supplied the poultry house with litter. After two weeks, the base of the walls and litter close to the walls appeared wet, dead chicks were being disposed off at the nearby rubbish heap. The labourers would be seen running in the poultry house to catch the birds by their legs for daily/routine practices.

The birds would be seen wings spreading and panting. The droppings would be scattered all over the poultry house. Janet supplied the birds to the local butcher for slaughtering. They used to hang the birds upside down by their legs before cutting their necks.

After slaughtering, the partially digested food would be found in the meat, crop and intestines. After slaughtering, less meat would be obtained. The buyers of the meat would complain about the quality of the meat.

Task

Explain how Janet should improve her project and production of good quality meat.

SECTION C
CROP PRODUCTION

ITEM 5

Mr Opio acquired 10 hectares of land under customary arrangement in a semi-arid region.

The land had previously been used for continuous grazing. He cleared the vegetation by burning and ploughed the soil one before planting cassava stem cuttings which was obtained from local farmers in areas that receive high annual rainfall. Two months after planting, foxtail grew fatter than the cassava plants between rows.

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Crop residues were removed after harvest, and fertilizers was applied occasionally without prior soil testing.

The crop relied only on rainfall. During dry spell, thin whitish patches appeared on some parts of soil surface, water lingered briefly after rainfall. Cassava plants developed curled leaves, small stems with nearly fused internodes and reduced tuber size. At harvest, some root tubers were pithy while minor cuts and bruises were observed on tubers. At the beginning of each planting season, Opio invited agricultural scientist to test the soil for electrical conductivity, and exchangeable sodium (Na %)

The results and cassava yield over five seasons are shown in table 1

Season	Ec/ds/m	Ph	Exchangeable (Na %)	Yield/dt/ha
1	2.1	6.8	4	22.5
2	3.4	7.2	7	18.0
3	5.6	7.8	11	13.2
4	7.6	8.3	16	8.4
5	10.5	8.9	22	4.6

Task

Explain how Mr opio can improve the productivity of his cassava project.

ITEM 6

James established a pineapple plantation of 4 hectares of land on steep slopes, the lower part had marshy vegetation, while the other dried quickly and developed cracks on the surface during the dry periods.

The land was cleared using surface tillage before planting pineapple suckers obtained from local neighbouring farms, three months after planting, some plants appeared closely packed, while a mixture of maize, sorghum and bean plants grew vigorously in the garden.

In some parts of the field, the soil surface was bare and exposed, whereas in others it was covered with shade and plants were leaning towards sunlight.

On several plants, leaves showed small holes and notched margins, while others had chewed leaf tips, frass deposits and yellowing edge near the base.

During harvesting, some fruits had split eyes and strong fermented smell while others had deep knife cuts, torn stalk attachments, and crushed crowns. James is concerned about the performance of his pineapple crop after investing a lot of money in the project.

Task

Suggest how the productivity of Mr. Jame's pineapple project can be improved.

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